

Missiles and data: A data-driven look into the patterns of Iran's missile firing during the war

Description: The recent conflict between Israel and Iran has been defined by direct acts of military aggression, notably characterized by significant missile exchanges and heightened geopolitical tensions. We present a data-driven investigation into the conflict, aiming to address complex research questions regarding the tactical and geopolitical dynamics of the war. Broadly, the study explores whether hidden strategic patterns stand at the base of the attacks, such as the deliberate synchronization of missile strikes on geographically distant targets. We also examine how these alert patterns changed over time during the war, and examine the intersection of political rhetoric and battlefield realities, seeking to determine if a statistical correlation exists between public statements by global leaders—specifically Donald Trump—and the volume and timing of Iranian missile launches. Through this dual analysis, the research seeks to shed light on covert attack strategies and the broader factors influencing acts of aggression during the conflict.

This project will aim to investigate these questions:

1. Is there a correlation between different locations that we wouldn't have predicted based on first impression?
2. How did the alert patterns change during the war?
3. Do the statements delivered by leaders have an actual effect on the aggression from the other side?

datasets:

Israel Home Front Command (HFC) Alerts Data: A historical log of security notifications and alarms issued by Israel's Home Front Command, continuously updated as new alerts are issued.

Size: The CSV file (israel-alerts.csv) is approximately 62.6 MB and contains tens of thousands of records spanning several years.

Fields: The raw dataset consists of columns such as **alertDate** (timestamp), **data** (comma-separated locations/cities where the alarm sounded), **category**.

The dataset includes all types of alerts (e.g., earthquakes, hostile aircraft, false alarms), not just missiles; also, the dataset holds information of alerts from 2014 and not only from the Iran war. Hence, we explicitly filtered the dataset to isolate only the dates of the Iran war and filtered unwanted alerts for the different experiments.

Throughout the alert analyses, we use the term “second-war dataset” for the HFC records from February 28 to June 17, 2026. We chose this period because both alert questions focus on this war and it contains enough alerts for the network and time-based analyses.

Official IDF Incoming-Threat Reports: Public messages from the IDF's official Telegram channel, used to identify reported launch origins and define the primary Iran-reporting phase.

Size: 2,311 official messages from February 28 to June 17, 2026, from which we extracted 416 candidate incoming-threat events.

Fields: Post time, reported launch origin, threat type, referenced alert time, timing basis, and source URL.

These reports do not cover every alert wave. We matched official source evidence to 345 of 1,196 reconstructed waves (28.8%). Therefore, we used the reports to provide context and define the analysis period, not as complete ground truth labels for every wave.

Donald Trump's Social Media Posts: A database of posts ("Truths" from Trump's own social network) written by Donald Trump, which we use to analyze the correlation between political rhetoric and acts of aggression.

- **Size:** 1,468 text posts dating from February 28, 2026, to June 17, 2026.

Our Initial attempts to independently scrape X (Twitter) and Truth Social using tools like Apify and Scweet were blocked by Cloudflare's security. Hence, we bypassed scraping blockers by using CNN's pre-compiled database of Truth's instead.

Evaluating the clustering of Home Front Command alarm logs and search for tactics:

Our primary objective was to investigate whether there is a spatial-temporal correlation between missile targets. Specifically, we aimed to determine if there is a deliberate tactical pattern of launching simultaneous attacks at two geographically distinct locations.

To evaluate this, we modeled the "ירי רקטות וטילים" (Rocket and Missile) alert logs as a temporal co-occurrence network graph.

- **Method:** We applied the **Girvan-Newman community detection algorithm** to identify distinct clusters of cities that consistently experience alerts simultaneously. Then, we searched for anomalies by distance, meaning we found two locations that were aimed simultaneously but it can not be from the same rocket because of their vast distance from each other.
- **Parameters:** We defined edges between cities if they triggered an alarm within the exact same **2-minute sliding window**. We isolate only the strongest synchronized events, and we applied a strict **edge weight threshold of > 70**.
It means only if an alarm was triggered within the exact same 2-minute sliding window More than 70 times, then we define an edge.
- **Modifications:** To prove the anomaly, we extracted the geographic coordinates of the most extreme outliers within a detected community to calculate the exact physical distance (in kilometers) between them, testing if their grouping was tactically viable for standard localized rocket fire.

Evaluation

The success of Girvan-Newman community algorithm was defined by the algorithm's ability to cluster cities into geographically logical and distinct zones (e.g., Western Galilee, Gush Dan, Negev) using *only* temporal co-occurrence data, without the model having access to actual GPS coordinates. We ensured our findings were not coincidental by measuring the maximum internal geographic distance within the detected clusters. If they all were close and in the same area of the country it means the algorithm was able to cluster correctly.

For the anomalies we calculated the distance between them and their cluster to identify them far away and caused by a simultaneous attack.

- **Setup:** The experiment isolated direct rocket alerts, built the 2-minute sliding window adjacency matrix, and clustered nodes with > 70 co-occurrences. To test the anomaly, we targeted the largest detected cluster (Community 5) and pulled geographic coordinates for its most extreme edges: the deep south (Ashdod), deep Samaria (Har Bracha/Yitzhar), the northern Sharon (Rishpon), and the eastern corridor (Jerusalem).

Results

The algorithm's output successfully passed our validation and sanity checks, yielding a network graph composed of nine distinct communities. The largest of these is the central cluster, comprising 367 cities predominantly located in the country's central region.

The results are best visualized using an interactive, force-directed network layout. In this visualization, Community 5 appears as a massive, dense central "hairball" dominating the graph.. This sharply contrasts with the tactical clusters (like the Northern border), which appear as small, isolated satellite islands on the periphery of the graph. Just creating a simple graph would have been not possible to understand because of the massive points so, we created two visualizations for an easy observation on the data:

The points scattered on Google maps:

https://www.google.com/maps/d/edit?mid=1oMMHxk_95npqDzXu1QVjL4rNidkbtS0&usp=sharing

Raw Girvan-Newman community graph. The graph is interactive, meaning you can zoom in and see the name of the city.

https://raw.githack.com/rankrausz/needle-project/refs/heads/main/CommunityAlerts/missiles%20-%20second%20war/graph_full.html?token=GHSAT0AAAAAAEETAIDEBDRWHEFFK6JA7NTK2UECVQQ

Tactics learned from anomaly:

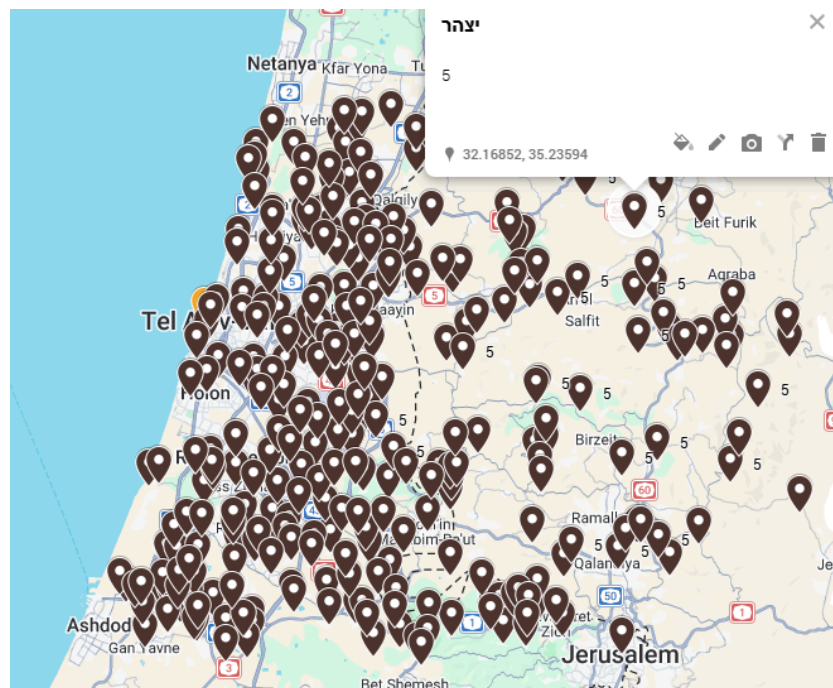
1. Normally, a direct "ירי רקטות וטילים" (Rocket and Missile) alert is highly tactical and localized. However, Community 5 (centered around גוש דן, but not bound to it) contains

367 cities, making up 73.1% of your entire network.

If you look closely at the cities within this single community, you will find extreme geographic far points in the exact same 2-minute window. For example:

- The Deep South & Coast: 'אשדוד - אזור תעשייה צפוני' (Ashdod) and 'קבוצת יבנה' (Kvutzat Yavne).
- Deep Samaria (Shomron): 'יצהר' (Yitzhar), 'איתמר' (Itamar), and 'הר ברכה' (Har Bracha).
- The Northern Sharon: 'רשפון' (Rishpon) and 'תל מונד' (Tel Mond).
- The Eastern Corridor: 'ירושלים' (Jerusalem) and 'מבשרת ציון' (Mevasseret Zion).

The distance between Ashdod on the southern coast and Yitzhar in the Shomron is over 70 kilometers. It is mathematically and physically impossible for a standard tactical rocket to trigger a continuous alert polygon across this entire span. This anomaly proves that it is a synchronized event of multiple missiles being shot to distinct areas consistently.



2. The algorithm formed an isolated and specific cluster (community 6) in the south. The community contains: 'נבטים' (Nevatim), 'באר שבע - מזרח' (Beer Sheva), 'עומר' (Omer), 'שב' (Segev Shalom), and 'תל ערד' (Tel Arad).



The clustering connected the Nevatim Airbase, Beer Sheva and its immediate surrounding Bedouin diaspora from the rest of the country with the Nevatim base as the geographic middle-point. Because it has no edges connecting it to the massive Community 5 (גוש דן), this supports the idea that there were Iran attacks aimed at Nevatim were a distinct, synchronized cluster of the barrage, completely separate from the missiles aimed at the center of the country.

How alert-wave structure changed during the second war

The network analysis above identifies localities that repeatedly appear together, but not how that structure changed over time. We therefore examined two additional aspects: the number of localities in each alert wave (a group of alerts occurring close together in time) and the overlap between consecutive waves. We asked whether waves included fewer localities over time while remaining frequent, and whether consecutive waves revisited the same localities. We call the number of distinct localities in a wave its alert footprint, it does not measure missile count, geographic area, or exposed population.

Method and evaluation

We first filtered only missile alerts (category = 1) and removed duplicate locality-timestamp rows. We reconstructed alert waves by grouping consecutive alerts when the gap between them was 8 minutes or less. This cutoff allows alerts from the same event to occur several minutes apart, while reducing the chance of merging separate events (we later repeated the analysis using gaps from 3 to 15 minutes). This produced 1,196 waves from 93,872 locality records.

The footprint distribution was highly skewed: most waves included few localities, while a small number included hundreds. We therefore applied a $\log(1 + \text{footprint})$ transformation and used K-means++ to find groups of waves with similar footprint sizes. We compared solutions with $k = 1, \dots, 12$ using relative cluster validation. To avoid treating small groups of unusual waves as meaningful families, we considered only solutions in which every cluster contained at least 5% of the waves. Among the valid solutions ($k = 2, \dots, 7$), the main SSE elbow occurred at $k = 2$, with an 84.8% reduction compared with $k = 1$. It also had the highest silhouette score, 0.765. The

clustering was stable across 20 separate K-means++ runs, with a mean pairwise adjusted Rand index of 0.997. The final result contained 853 localized waves with 1–25 localities and 343 broad waves with at least 26 localities.

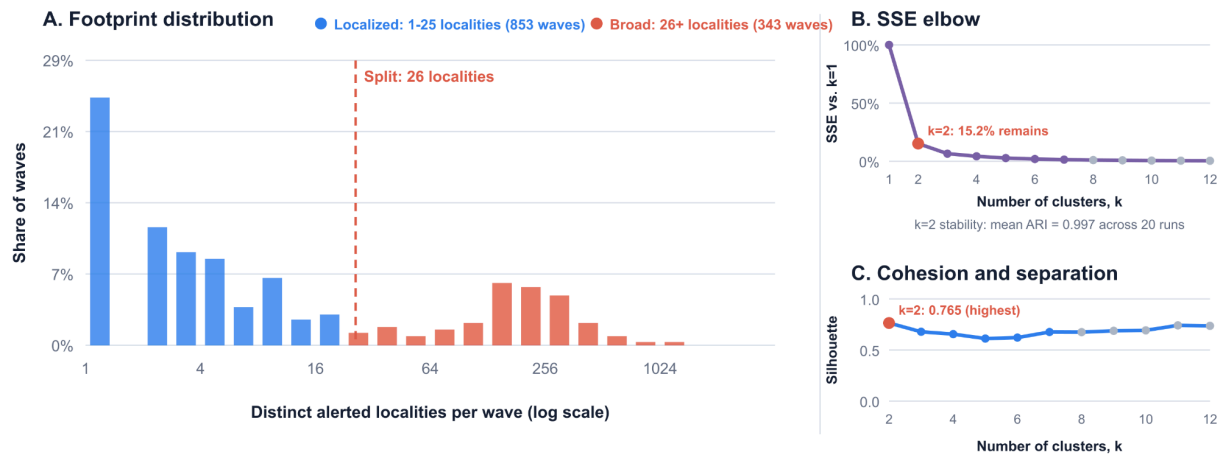


Figure 1. Alert-wave footprint families and relative validation. The log scaled distribution, SSE elbow, and silhouette comparison support the data-derived 26-locality boundary.

Among waves with timing-matched official source labels, 258 of 263 broad waves (98.1%) were Iranian, while 44 of 49 Lebanese waves (89.8%) were localized (Fisher's exact odds ratio = 63.1, $p < 0.001$). However, only 55.0% of labeled localized waves were Lebanese. Thus, broad footprint strongly indicated Iran, but localized footprint did not uniquely indicate the northern front.

We defined the primary phase as February 28 to April 8, 2026 - the longest consecutive period with official reports of threats from Iran. We aggregated waves by day and used Spearman correlation to test the relationship between phase day and daily mean footprint. For short-term persistence, we calculated Jaccard similarity between consecutive same-day waves and compared the observed mean with a null distribution from 2,000 within day order shuffles. The shuffling preserved each wave's footprint and localities but randomized adjacency.

Results

During the 40-day primary phase, daily mean footprint declined ($\rho = -0.60$, $p < 0.001$). It decreased from 212.5 localities per wave in the first seven days to 61.7 in the last seven - a 71% reduction (one sided Mann–Whitney $p = 0.0055$). Wave counts showed no downward trend ($\rho = +0.21$, $p = 0.19$), averaging 19.1 and 22.3 waves per day in the first and last seven days. The daily median footprint of broad waves also declined ($\rho = -0.50$, $p = 0.0011$), showing that the contraction was not caused only by a shift toward localized waves.

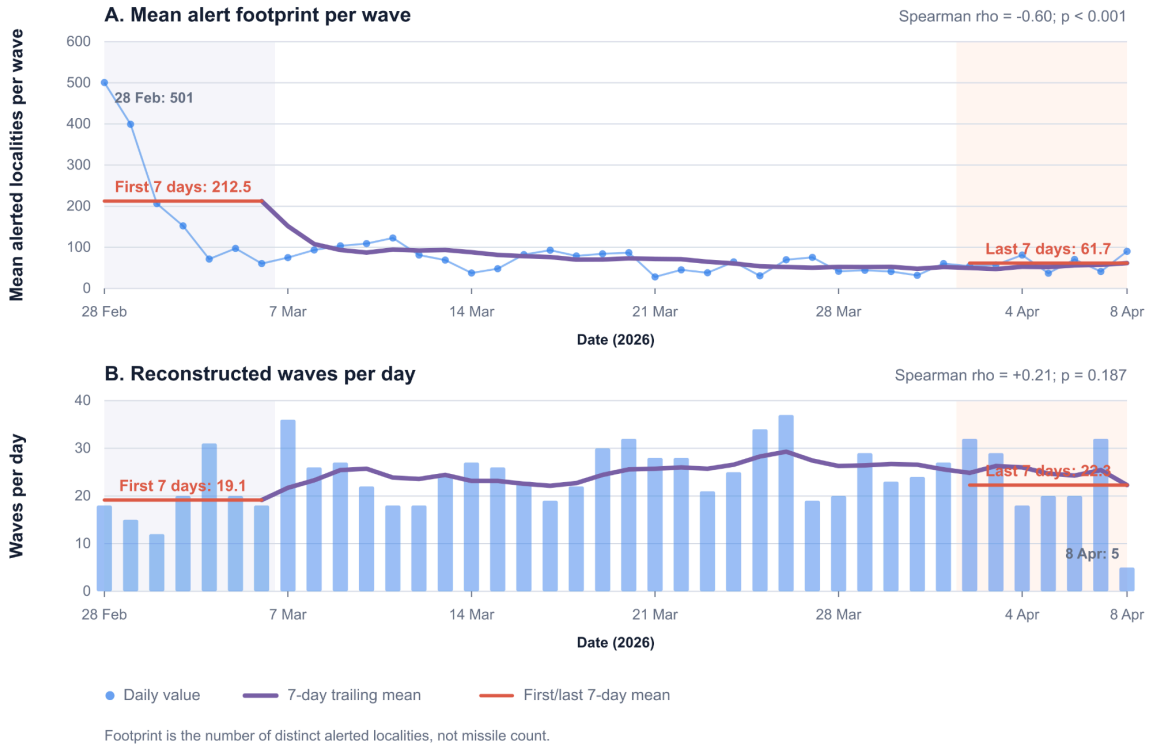
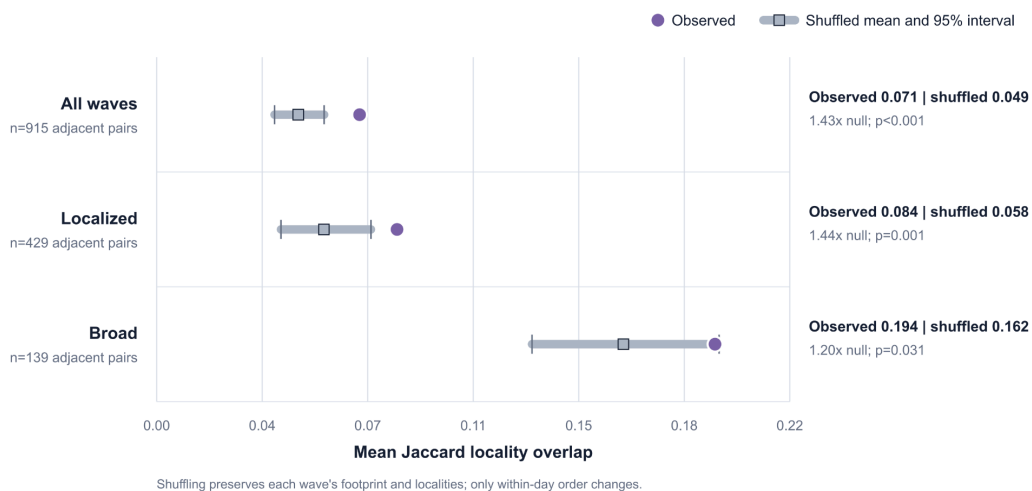


Figure 2. Footprint contraction while waves remained frequent. Daily observations, seven day trailing means, and directly labeled endpoint averages show the two quantities on separate, zero-based axes.

Across 915 consecutive same-day wave pairs, the observed mean Jaccard similarity was 0.071, compared with 0.049 under the shuffled null model - difference of 0.021, with the observed overlap being 1.43 times the null expectation (permutation p < 0.001). The result therefore provides some evidence of short term locality persistence, although the effect itself is small. The result for localized wave pairs remained consistent across the tested wave-gap definitions. The evidence for broad wave pairs was weaker and was not distinguishable from the null when the longest, 15-minute gap was used.

Figure 3. Short-term locality persistence. Observed overlap is compared directly with the mean and central 95% interval of the shuffled baseline.



Robustness, limitations, and conclusion

The footprint trend remained negative after excluding the unusually broad first seven days ($\rho = -0.44$, $p = 0.011$), across wave gaps from 3 to 15 minutes, and after shifting the phase endpoint. This shows that the result did not depend on the opening days or a single parameter choice.

The time-gap rule can join separate events into one wave, the longest reconstructed wave lasted 42 minutes. Official source labels covered only 28.8% of waves, and coverage was uneven: 77.3% of broad waves but only 9.4% of localized waves received a label. The source family result therefore applies only to the labeled subset. In addition, the high-confidence labeled subset did not show a clear temporal footprint trend ($\rho = -0.13$, $p = 0.44$). The locality overlap may also partly reflect consecutive waves from the same front. Therefore, these observational results cannot establish changes in launch numbers, military strategy, or defense performance.

Overall, waves remained frequent while their recorded footprint decreased. Among source-labeled waves, broad footprint was strongly associated with Iran, while localized waves had mixed sources. Consecutive waves also showed modest locality overlap.

Correlation between leaders statements and acts of aggression:

During the war, we were sitting in shelters watching TV commentators excessively discussing individual statements from Donald Trump & Benjamin Netanyahu. We felt there was no evidence for the impact of every single statement and speech delivered on the course of the war. Therefore, we set out to see whether there was a correlation between statements from leaders during the war, and missile-firing from Iran to Israel.

We started by crawling and scraping Trump's ("@realDonaldTrump") X account. We tried a couple of tools like the `Apify` and `Scweet`, but X was shut down pretty efficiently. Hence, we turned to D. Trump's own network "Truth" - which was also locked down by cloudflare, which blocked all our scraping attempts. Luckily, we found out CNN [keeps track](#) of all his tweets and we were able to use this updating database.

Setup: We used 1468 Trump "truths" dating from 28.2.2026 to 17.6.2026. On every "truth" text, we extracted an embedding vector of size 384x1 using `all-MiniLM-L6-v2`.

To test the reliability of the embedding model we chose, we measured the cosine similarity between the representation of the beloved phrase "fake news" and each vector. The text that measured closest to "fake news" started with the sentence *"Content snippet: No one can believe that Fake News CNN put out a knowingly false and dangerous statement pretending it came from the upper levels of the Iranian Government"*.

To further verify the embeddings capture war-related content, and more specifically threats that might cause Iran to use aggression, we performed K-means++ clustering. We wanted to try 2 to 20 centroids with cosine similarity as a distance metric; we chose cosine similarity to measure the whole direction of a vector and not a pairwise entry distance such as in L1. We wanted to use sklearn's K-means++, but it did not support cosine similarity.

We will make a note that for 2 unit vectors A, B it holds that:

$$|A| = |B| = 1 \rightarrow ||A - B||^2 = |A|^2 + |B|^2 - 2AB = 2 - 2AB$$

And:

$$|A| = |B| = 1 \rightarrow 1 - \frac{A \cdot B}{|A| \times |B|} = 1 - \frac{A \cdot B}{1 \times 1} = 1 - AB$$

So minimizing cosine similarity for A,B is equivalent to minimizing the euclidean distance. Hence, we normalized all embedding vectors and used sklearn's K-means++ with L2 as the distance function.

The average distances from points to their centroids seemed to start drop significantly slower starting from K=10, so we printed 5 sentences from each cluster for K=10. Two of the clusters seemed very much war-related, with sentences like:

"A Country supplying Military Weapons to Iran will be immediately tarified, on any and all goods sold to the United States of America, 50%, effective immediately. There will be no exclusions or exemptions! President DJT"

In one, And:

"Watch Mark Levin interview of Brilliant Marc Thiessen tonight at 8:00 P.M., on FoxNews. Will discuss the importance of hitting Iran, HARD!!! President DJT"

In the other; while other clusters seemed to focus on other topics like economy and applauding other politicians/world leaders. That led us to think that the `all-MiniLM-L6-v2` is indeed expressive enough; and that if there is a connection between aggressive statements from president Trump and Iran "retaliating" to these statements - we can detect it using the embedding vectors.

Experiment:

In order to check the correlation itself, we used Distance Matrix Correlation (Mantel Test): We calculated the cosine similarity between every pair of "truths" and the point-wise difference of the missile volume graph of the following 24 hours, to see if similar truths match to similar missile volume graphs in the following 24 hours. The missile volume graph had a datapoint for every 10 minutes, totaling 144 datapoints.

The p-value measured was smaller than 0.0001, showing statistical significance, and the r-value resulted in 0.053. To interpret our results, we turned to Qu and Kazakov (2016)¹, who utilized an identical Representational Similarity Analysis pipeline - correlating text embedding distance matrices against time-series distance matrices using the Mantel Test.

In their analysis of financial news embedding (extracted by word2vec) against stock-market data, they reported Mantel correlations ranging from $r = 0.0149$ to $r = 0.1998$ (Qu and Kazakov 2016, Table II). Our finding of $r = 0.0530$ falls squarely within this range. However, while the

¹ Stefanov, Stefan, and Daniel Kudenko. "Quantifying correlation between financial news and stocks." In *2016 IEEE Symposium Series on Computational Intelligence (SSCI)*, pp. 1-8. IEEE, 2016.

embedding correlations in their financial baseline mostly did not achieve statistical significance (e.g., $p > 0.40$), our correlation achieved a highly significant $p < 0.0001$.

This suggests that while the effect of Trump's statements on "Truth" on Iranian launches is predictably small, the link is "at least" as reliable as the link between Trump's statements and the stock market prices. It is common to believe that the relation between politician's statements and the stock exchange is undeniable, so we find it easy to believe that the Iranian firing was similarly affected by political statements.

